Permutations

1

a 120 **b** 42 **c** 120 **d** 120

e 1 **f** 11 **g** 210 **h** 20 160

2 362 880

3 120

4 120

Probabilities with permutations

5 $\frac{1}{2}$

6

a 24 **b** 6 **c** $\frac{1}{4}$

7

a $\frac{5}{6}$ **b** 72

8

a $\frac{3}{28}$ **b** $\frac{15}{28}$

9

a 120 **b** $\frac{1}{5}$ **c** $\frac{1}{120}$

10

a 40 320 **b** $\frac{1}{40\,320}$

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11

a $\frac{1}{3}$ b $\frac{2}{3}$ c 0 d $\frac{1}{5}$

e $\frac{4}{5}$

12

a $\frac{1}{120}$ b $\frac{1}{120}$ c $\frac{1}{120}$ d $\frac{2}{5}$

e $\frac{3}{5}$ f $\frac{3}{5}$

Applications

13 120

14 5040

15

a $\frac{1}{120}$ b $\frac{1}{120}$ c $\frac{1}{120}$ d $\frac{1}{5}$

16

a $17\,576\,000$ b $\frac{1}{17\,576\,000}$ c $\frac{1}{13}$

17

a 720 b 120 c $\frac{1}{6}$

18

a $\frac{1}{90}$ b $\frac{1}{90}$ c $\frac{1}{45}$ d $\frac{4}{9}$

19 $\frac{1}{2\,793\,510\,720}$

20 $\frac{7}{11}$

21

a $\frac{1}{}$ b $\frac{29\,889}{}$

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22 **a** $\frac{1}{210}$ **b** $\frac{1}{35}$

23 **a** $\frac{1}{156}$ **b** $\frac{1}{1716}$

24 $\frac{1}{720}$

25 $\frac{1}{22}$

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